

1.5 LONG-TERM PROJECTS

Program Theme

Digital Wellness, Human Behavior, and Technology

Computer Science Pathway

Digital Wellness Analytics Platform

Community Partner: Neuroscience & Psychology Club

Students develop a web-based analytics platform that investigates the relationship between technology usage habits and student wellness. Students create anonymous survey instruments, collect voluntary anonymous survey data, organize and analyze datasets, develop visualizations, and publish findings through an interactive website dashboard.

Deliverables

- Project Proposal
- Anonymous Survey Instrument
- Dataset
- Data Analysis Report
- Interactive Website Dashboard
- Final Presentation

Total Time

23 Hours

Industry Skills

- Data Collection

- Data Cleaning
- Data Visualization
- Front-End Web Development
- User Experience Design
- Software Testing
- Project Documentation

Georgia Standards Alignment

Introduction to Software Technology (IT-FND-4, IT-FND-5, IT-FND-6)

- Analyze and interpret data.
- Develop technology solutions to solve real-world problems.
- Apply web development and user interface design principles.
- Demonstrate project planning and documentation skills.
- Communicate technical findings to stakeholders.

AP Computer Science Principles

- Big Idea 1: Creative Development
- Big Idea 2: Data
- Big Idea 3: Algorithms and Programming
- Big Idea 5: Impact of Computing

Game Design Pathway

Digital Dopamine: Understanding Technology Through Interactive Play

Community Partner: Neuroscience & Psychology Club

Students develop a lightweight 2D Unity educational game that demonstrates how technology habits influence focus, attention, reward systems, habit formation, and digital wellness. The game emphasizes simple assets, scene transitions, educational gameplay mechanics, and accessibility within school hardware constraints.

Deliverables

- Game Design Document
- Storyboard
- Asset Library
- Playable Unity Prototype

- Testing Report
- Final Presentation

Total Time

22 Hours

Industry Skills

- Game Design
- Unity Development
- C# Programming
- User Interface Design
- Debugging
- Asset Management
- Interactive Storytelling

Georgia Standards Alignment

Game Design & Simulation Standards

- GDAS1: Explore careers and opportunities in game development.
- GDAS2: Apply principles of game design and development.
- GDAS3: Create digital assets and game components.
- GDAS4: Develop and test interactive games.
- GDAS5: Evaluate, revise, and present completed projects.

Computer Science Practices

- Computational Thinking
- Iterative Design
- Problem Solving
- Collaboration
- Software Testing

Blank Long-Term Project Rubric

Criteria	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)	Score
Project Planning					

Research & Documentation					
Technical Implementation					
Creativity & Innovation					
Team Collaboration					
Testing & Revision					
Communication & Presentation					
Professionalism					

Total Points: ____ / 32

Teacher Comments

Student Team List

Computer Science Pathway Teams

Team #	Team Name	Student 1	Student 2	Student 3	Student 4	Teacher Approval
1						
2						

Game Design Pathway Teams

Team #	Team Name	Student 1	Student 2	Student 3	Student 4	Teacher Approval
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1						
2						

Project Approval Form

Team Name: _____

Project Title: _____

Pathway: ☐ Computer Science ☐ Game Design

Community Impact Statement:

Teacher/Consultant Approval: _____

Approval Date: _____

This addition makes Section 1.5 audit-ready by showing standards alignment, assessment methodology, student collaboration documentation, and approval procedures.